



NORTH SYDNEY BOYS HIGH SCHOOL

MATHEMATICS (EXTENSION 1)

2015 HSC Course Assessment Task 4

Friday August 14, 2015

General instructions

- Working time – 55 min.
(plus 5 minutes reading time)
- **Commence each new question on a new page.**
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets used in the correct order within this paper and hand to examination supervisors.

Class (please ✓)

- ☐ 12M3A – Ms Ziazaris
- ☐ 12M3B – Mr Berry
- ☐ 12M3C – Mr Zuber
- ☐ 12M4A – Mr Weiss
- ☐ 12M4B – Mr Lin
- ☐ 12M4C – Mr Ireland

STUDENT NUMBER **# BOOKLETS USED:**

Marker's use only.

QUESTION	1	2	3	4	Total	%
MARKS	$\overline{8}$	$\overline{12}$	$\overline{9}$	$\overline{8}$	$\overline{37}$	

- Question 1** (8 Marks) Commence a NEW page. **Marks**
- (a) Find the constant term in the expression of $\left(x - \frac{1}{2x^3}\right)^{20}$. **3**
- (b) Prove that $\binom{n+1}{r} = \binom{n}{r-1} + \binom{n}{r}$. **2**
- (c) In the expression of $(2 + 3x)^n$, the coefficients of x^3 and x^4 are in the ratio 8 : 15. **3**
- Find the value of n .

- Question 2** (12 Marks) Commence a NEW page. **Marks**
- (a) A cat is sitting on the top of a wall measuring 3.2 m high. It sees a mouse on the ground 4 m away from the foot of the wall. The cat jumps horizontally from the top of the wall at an initial speed of 6 ms^{-1} . Take $g = 10 \text{ ms}^{-2}$. **2**
- i. Show that the equations of motion are
- $$\begin{cases} x = 6t \\ y = -5t^2 + 3.2 \end{cases}$$
- ii. How long does it take the cat to reach the ground? **1**
- iii. Show that the cat misses the mouse. **1**
- iv. If the cat is to land on the mouse, calculate the speed at which the cat should leap off the wall. **1**
- (b) A projectile is fired from a point P on horizontal ground with initial speed $V \text{ ms}^{-1}$ at an angle of 45° . The only force acting on the projectile is due to gravity, measuring $g \text{ ms}^{-2}$. **1**
- i. State the equations of motion. **1**
- ii. Determine the coordinates (X, Y) of the point at its greatest height. **2**
- iii. Prove that a second projectile, fired from P with an initial speed $V \text{ ms}^{-1}$, at an angle of elevation of θ , where $\tan \theta = 3$, will also pass through (X, Y) . **2**
- iv. If the projectiles are fired simultaneously, what are the coordinates of the second projectile at the instant the first projectile lands? **2**

Question 3 (9 Marks)

Commence a NEW page.

Marks

- (a) A particle moves along the x axis. Its velocity is given by

$$v = \pm\sqrt{8x - x^2}$$

- i. Prove the particle is undergoing simple harmonic motion. **2**
 - ii. Find the acceleration of the particle when $x = 3$. **1**
 - iii. Find the amplitude of the motion. **2**
- (b) Assuming that tides rise and fall in simple harmonic motion, a ship needs 11 m of water to pass through a channel safely. At low tide, the channel is 8 m deep and at high tide it is 12 m. Low tide occurs at 10 am and high tide occurs at 4 pm. **4**

Find the period of time for which the ship can safely proceed through the channel.

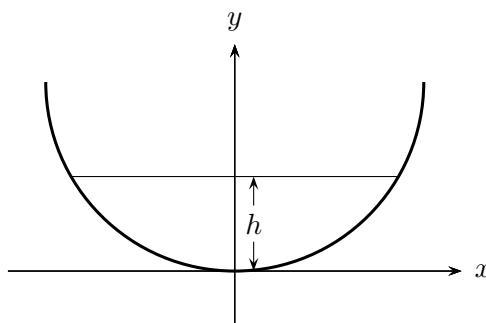
Question 4 (8 Marks)

Commence a NEW page.

Marks

The diagram shows the lower half of the circle whose equation is

$$x^2 + y^2 - 20y = 0$$



A hemispherical bowl is obtained by rotating the semicircle about the y axis.

- (a) Show that when the depth of water is h cm:

- i. The volume of water V is given by **2**

$$V = \frac{\pi h^2}{3}(30 - h) \text{ cm}^3$$

- ii. The surface area of water exposed to the air is given by **2**

$$S = \pi h(20 - h) \text{ cm}^2$$

- (b) If water is poured in at $50 \text{ cm}^3/\text{s}$, determine the following when the depth of water is 5 cm:

- i. the rate of increase of the depth of water. **2**
- ii. the rate of increase of the surface area of the water. **2**

End of paper.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1} + C, \quad n \neq -1; \quad x \neq 0 \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x + C, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax} + C, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax + C, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax + C, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax + C, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax + C, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C, \quad a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right) + C, \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right) + C$$

NOTE: $\ln x = \log_e x$, $x > 0$

Q1.

$$(a) \left(x - \frac{1}{2x^3}\right)^{20} = \sum_{k=0}^{20} \binom{20}{k} x^{20-k} \left(-\frac{1}{2}\right)^k x^{-3k}$$

$$= \sum_{k=0}^{20} \binom{20}{k} x^{20-4k} \left(-\frac{1}{2}\right)^k$$

x^0 term when $20-4k=0$
ie $k=5$

✓

$$\text{Term is: } \binom{20}{5} \left(-\frac{1}{2}\right)^5 = -\frac{969}{2}$$

✓

$$(b) {}^nC_r + {}^nC_{r-1} = \frac{n!}{r!(n-r)!} + \frac{n!}{(r-1)!(n-r+1)!}$$

$$= \frac{n!(n-r+1)}{r!(n-r+1)!} + \frac{r \cdot n!}{r!(n-r+1)!}$$

✓

$$= \frac{n!(n-r+1+r)}{r!(n-r+1)!}$$

$$= \frac{(n+1)!}{r!(n-r+1)!}$$

✓

$$= {}^{n+1}C_r$$

OR: use coefficients of $(1+x)^{n+1} = (1+x)(1+x)^n$

LHS coeff on ~~x^r~~ x^r is ${}^{n+1}C_r$

RHS coeff of x^r is $({}^nC_{r-1})x^{r-1} \times x + 1 \cdot {}^nC_r x^r$

$$= {}^nC_{r-1} + {}^nC_r$$

$$\therefore {}^{n+1}C_r = {}^nC_{r-1} + {}^nC_r$$

Q1 (c) Ratio coefficients :

$$\frac{{}^nC_3 2^{n-3} 3^3}{{}^nC_4 2^{n-4} 3^4} = \frac{8}{15}$$

✓

$$\frac{{}^nC_3}{{}^nC_4} \cdot \frac{2}{3} = \frac{8}{15}$$

$$\frac{{}^nC_3}{{}^nC_4} = \frac{24}{30}$$

$$\frac{4}{n-3} = \frac{4}{5}$$

$$\therefore n = 8$$

Question 2

(a)(i) $\ddot{x} = 0$
 $\dot{x} = c$
 $\dot{x} = 6$ (horizontal)
 $x = \int 6 dt$
 $x = 6t + c$

when $t = 0$ $x = 0$
 $\therefore x = 6t$

$\ddot{y} = -10$
 $\dot{y} = \int -10 dt$
 $\dot{y} = -10t + c$
 when $t = 0$ $\dot{y} = 0$
 $0 = -10(0) + c$
 $c = 0$
 $\dot{y} = -10t$
 $y = \int -10t dt$
 $y = -5t^2 + c$
 when $t = 0$ $y = 3.2$
 $3.2 = -5(0) + c$
 $\therefore c = 3.2$
 $y = -5t^2 + 3.2$

(ii) when $y = 0$
 $0 = -5t^2 + 3.2$
 $5t^2 = 3.2$
 $t^2 = \frac{3.2}{5}$
 $t = \pm \sqrt{\frac{3.2}{5}}$
 $\therefore t = 0.8$ ($t > 0$)

(iii) when $t = 0.8$

$$x = 6(0.8)$$

$$x = 4.8 \text{ m}$$

$\therefore$ the cat lands at $x = 4.8$
 and misses the mouse
 which is at $x = 4 \text{ m}$.

(iv) speed = $\frac{\text{distance}}{\text{time}}$
 $= 4 \div (0.8)$
 $= 5 \text{ m/s}$

(b)(i) $\ddot{x} = 0$
 $\dot{x} = V \cos \theta$
 $x = Vt \cos \theta$

$\ddot{y} = -g$
 $\dot{y} = -gt + V \sin \theta$
 $y = -\frac{1}{2}gt^2 + Vt \sin \theta$

since $\sin 45^\circ = \frac{1}{\sqrt{2}}$ and
 $\cos 45^\circ = \frac{1}{\sqrt{2}}$

$\ddot{x} = 0$
 $\dot{x} = \frac{V}{\sqrt{2}}$
 $x = \frac{Vt}{\sqrt{2}}$

$\ddot{y} = -g$
 $\dot{y} = -gt + \frac{V}{\sqrt{2}}$
 $y = -\frac{1}{2}gt^2 + \frac{Vt}{\sqrt{2}}$

(ii) when $y = 0$

$$-gt + \frac{v}{\sqrt{2}} = 0$$

$$gt = \frac{v}{\sqrt{2}}$$

$$t = \frac{v}{g\sqrt{2}} \dots \textcircled{1}$$

sub ① into x & y

$$x = \frac{vt}{\sqrt{2}}$$

$$x = \frac{v \left(\frac{v}{g\sqrt{2}} \right)}{\sqrt{2}}$$

$$x = \frac{v^2}{2g}$$

$$y = -\frac{1}{2}gt^2 + \frac{vt}{\sqrt{2}}$$

$$y = -\frac{1}{2}g \left(\frac{v}{g\sqrt{2}} \right)^2 + \frac{v \left(\frac{v}{g\sqrt{2}} \right)}{\sqrt{2}}$$

$$y = -\frac{v^2}{4g} + \frac{v^2}{2g}$$

$$y = \frac{v^2}{4g}$$

$\therefore$ coordinates are

$$\left(\frac{v^2}{2g}, \frac{v^2}{4g} \right)$$

(iii) equations of motion are:

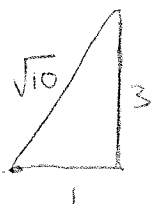
$$\ddot{x} = 0$$

$$\ddot{y} = -g$$

$$\dot{x} = v \cos \theta \quad \dot{y} = -gt + v \sin \theta$$

$$x = vt \cos \theta \quad y = -\frac{1}{2}gt^2 + vt \sin \theta$$

since $\tan \theta = 3$



$$\sin \theta = \frac{3}{\sqrt{10}}$$

$$\cos \theta = \frac{1}{\sqrt{10}}$$

$\therefore$ equations of motion are

$$\ddot{x} = 0$$

$$\dot{x} = \frac{v}{\sqrt{10}}$$

$$x = \frac{vt}{\sqrt{10}}$$

$$\ddot{y} = -g$$

$$y = -gt + \frac{3v}{\sqrt{10}}$$

$$y = -\frac{1}{2}gt^2 + \frac{3vt}{\sqrt{10}}$$

find when $x = \frac{v^2}{2g}$

$$\frac{vt}{\sqrt{10}} = \frac{v^2}{2g}$$

$$t = \frac{v\sqrt{10}}{2g}$$

show that when $t = \frac{v\sqrt{10}}{2g}$

that $y = \frac{v^2}{4g}$

$$y = -\frac{1}{2}gt^2 + \frac{3vt}{\sqrt{10}}$$

$$y = -\frac{1}{2}g \left(\frac{v\sqrt{10}}{2g} \right)^2 + \left(\frac{3v}{\sqrt{10}} \right) \left(\frac{v\sqrt{10}}{2g} \right)$$

$$= -\frac{10v^2}{8g} + \frac{3v^2}{2g}$$

$$= -\frac{5v^2}{4g} + \frac{6v^2}{4g}$$

$\frac{v^2}{4g}$, as required

$\therefore$ 2nd projectile passes through (x, y) since

when $x = \frac{v^2}{2g}$ $y = \frac{v^2}{4g}$

(iii) alternative solution.

$$x = Vt \cos \theta \quad \dots \textcircled{1}$$

$$y = -\frac{1}{2}gt^2 + Vt \sin \theta \quad \dots \textcircled{2}$$

rearrange $\textcircled{1}$ to get

$$t = \frac{x}{V \cos \theta}$$

and sub into $\textcircled{2}$

$$y = -\frac{1}{2}gt^2 + Vt \sin \theta$$

$$y = -\frac{1}{2}g\left(\frac{x}{V \cos \theta}\right)^2 + V\left(\frac{x}{V \cos \theta}\right) \sin \theta$$

$$y = \frac{-gx^2}{2V^2 \cos^2 \theta} + x \tan \theta$$

$$y = \frac{-gx^2}{2V^2} \sec^2 \theta + x \tan \theta$$

$$y = -\frac{gx^2}{2V^2} (\tan^2 \theta + 1) + x \tan \theta$$

we now have to show that $x = \frac{V^2}{2g}$, $y = \frac{V^2}{4g}$ & $\tan \theta = 3$ satisfy the cartesian equation above.

$$\text{LHS} = \frac{V^2}{4g}$$

$$\text{RHS} = -\frac{g}{2V^2} \left(\frac{V^2}{2g}\right)^2 (3^2 + 1) + \frac{V^2}{2g} (3)$$

$$= -\frac{g}{2 \times g^2} \frac{(10)^5}{g^2} + \frac{3V^2}{2g}$$

$$= -\frac{5V^2}{4g} + \frac{6V^2}{4g}$$

$$= \frac{V^2}{4g}$$

$$= \text{LHS}$$

the projectile passes through (x, y)

(iv) find t when $y = 0$ for the first projectile.

$$-\frac{1}{2}gt^2 + \frac{Vt}{\sqrt{2}} = 0$$

$$t \left(-\frac{1}{2}gt + \frac{V}{\sqrt{2}}\right) = 0$$

$$-\frac{1}{2}gt + \frac{V}{\sqrt{2}} = 0 \quad (t > 0)$$

$$\frac{1}{2}gt = \frac{V}{\sqrt{2}}$$

$$t = \frac{2V}{g\sqrt{2}}$$

sub into x & y for the 2nd projectile.

$$x = \frac{Vt}{\sqrt{10}}$$

$$x = \frac{V}{\sqrt{10}} \left(\frac{2V}{g\sqrt{2}}\right)$$

$$x = \frac{2V^2}{g\sqrt{20}}$$

$$x = \frac{V^2}{g\sqrt{5}}$$

$$y = -\frac{1}{2}gt^2 + \frac{3Vt}{\sqrt{10}}$$

$$y = -\frac{1}{2}g\left(\frac{2V}{g\sqrt{2}}\right)^2 + \frac{3V}{\sqrt{10}} \left(\frac{2V}{g\sqrt{2}}\right)$$

$$y = -\frac{4V^2}{2 \times g^2 \times 2} + \frac{6V^2}{g\sqrt{20}}$$

$$y = -\frac{V^2}{g} + \frac{6V^2}{2g\sqrt{5}}$$

$$y = \frac{3V^2 - \sqrt{5}V^2}{g\sqrt{5}}$$

$\therefore$ coordinates are

$$\left(\frac{V^2}{g\sqrt{5}}, \frac{V^2(3 - \sqrt{5})}{g\sqrt{5}}\right)$$

Question 3 EXT1 SOLN'S TASK 4 2015

a) (i) $v = \pm \sqrt{8x - x^2}$

$$v^2 = 8x - x^2$$

$$\frac{1}{2}v^2 = 4x - \frac{x^2}{2}$$

$$\ddot{x} = \frac{d}{dx} \left(\frac{1}{2}v^2 \right)$$

$$= \frac{d}{dx} \left(4x - \frac{x^2}{2} \right)$$

$$\therefore \ddot{x} = 4 - x$$

$$\ddot{x} = -1(x - 4)$$

which is in the form

$$\ddot{x} = -n^2(x - x_0) \text{ where } n=1, x_0 = \text{centre of motion is } 4.$$

$\therefore$ Particle in SHM.

(OR you could say it is proportional to the negative displacement)

(ii) $\ddot{x} = 4 - x$

When $x = 3$

$$\ddot{x} = 1$$

(iii) When $v = 0$.

$$0 = 8x - x^2$$

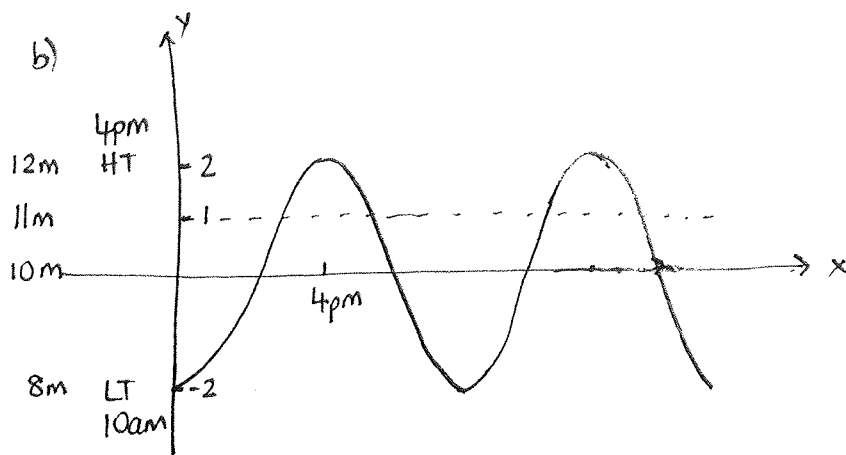
$$x = 0, x = 8.$$

Comes to rest at $x = 0$ $x = 8$

Centre of motion : $x = 4$

$$\begin{array}{c} | \quad | \quad | \\ 0 \quad 4 \quad 8 \\ \underbrace{\quad \quad} \end{array}$$

$\therefore$ Amplitude $a = 4$



Period = 12 hours

$$\frac{2\pi}{n} = 12$$

$$\therefore n = \frac{\pi}{6} \quad \text{Amplitude} = 2$$

$$x = -a \cos nt$$

$$x = -2 \cos \frac{\pi}{6} t$$

When 11m, $x = 1$

$$1 = -2 \cos \frac{\pi}{6} t$$

$$\cos \frac{\pi}{6} t = -\frac{1}{2}$$

$$\frac{\pi t}{6} = \frac{2\pi}{3}, \frac{4\pi}{3}$$

$$t = 4, 8$$

Will be above 11m 4h and 8h after 10am

$\therefore$ Can enter through 2pm and 6pm.

BUT in 24h period can also enter
between 2am - 6am.

OR $\rightarrow x = -a \cos nt + x_0$

$$x = -2 \cos \frac{\pi}{6} t + 10$$

When $x = 11$

$$11 = -2 \cos \frac{\pi}{6} t + 10$$

Q4. (a) (i) $V = \pi \int_0^h x^2 dy$

$\checkmark$ $= \pi \int_0^h (20y - y^2) dy$

$= \pi \left[10y^2 - \frac{y^3}{3} \right]_0^h$

$= \pi \left(10h^2 - \frac{h^3}{3} \right)$

$\checkmark$ $= \frac{\pi h^2}{3} (30 - h) \text{ cm}^3$

(b) $\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$ (1)

Now: $\frac{dV}{dt} = 50$ (given)

$V = \pi \left(10h^2 - \frac{1}{3}h^3 \right)$

$\frac{dV}{dh} = \pi (20h - h^2)$

At $h=5$

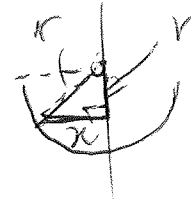
$\frac{dV}{dh} = \pi (20(5) - 25)$

$= 75\pi$

$\therefore 50 = 75\pi \cdot \frac{dh}{dt}$ from (1)

or $\frac{dh}{dt} = \frac{50}{75\pi}$

$= \frac{2}{3\pi} \text{ cm/sec}$

(ii)  $x^2 = r^2 - (r-h)^2$
 $r=10$
 $\therefore x^2 = 10^2 - (10-h)^2$
 $= 20h - h^2$

or $y=h: x^2 + h^2 - 20h = 0$ $\checkmark$
 $x^2 = 20h - h^2$

S.A. $= \pi x^2$

$= \pi (20h - h^2)$

$= \pi h (20 - h)$ as required. $\checkmark$

(ii) $\frac{dS}{dt} = \frac{dS}{dh} \cdot \frac{dh}{dt}$ (2)

$S = \pi (20h - h^2)$

$\frac{dS}{dh} = \pi (20 - 2h)$

At $h=5$, $\frac{dS}{dh} = 10\pi$

$\therefore \frac{dS}{dt} = 10\pi \cdot \frac{2}{3\pi}$ (from (i))

$= \frac{20}{3} \text{ km/s.}$